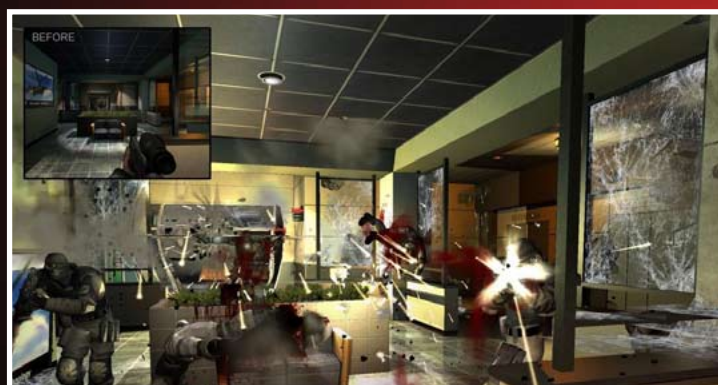




JUPITER EXTENDED TECHNOLOGY

OVERVIEW

Jupiter Extended ("EX") Technology is the latest multi-platform game development kit from Touchdown Entertainment. Driven by an all-new DirectX 9 renderer, Havok Game Dynamics and numerous other upgrades, Jupiter EX, is the ideal technology for teams creating advanced 3D games.



Jupiter EX is a full-featured game development system with visual technologies, Havok Physics, networking and a suite of powerful content creation tools. Jupiter EX also includes game code to the award-winning game F.E.A.R. The sample game code enables developers to leverage standard game-oriented functions and objects and focus on the unique characteristics of their Jupiter EX powered game.

FOCUS ON FUN

VISUAL TECHNOLOGIES

RENDERING

Jupiter EX is powered by a new flexible, extensible, and data driven DirectX 9 renderer that uses materials for rendering all visual objects. Each material associates an HLSL shader with artist-editable parameters used for rendering, including texture maps (normal, specular, emissive, etc.), colors, and numeric constants.

LIGHTING MODEL

Jupiter EX features a unified Blinn-Phong per-pixel lighting model, allowing each light to generate both diffuse and specular lighting consistently across all solid objects in the environment. The lighting pipeline uses the following passes:

Emissive

The emissive pass allows objects to display a glow effect and establishes the depth buffer to improve performance.

Lighting

The lighting pass renders each light, first by generating shadows and then by applying the lighting onto any pixels that are visible and not shadowed.

Translucency

The translucent pass blends all translucent objects into the scene using back to front sorting.



SAMPLE LIGHTS

Jupiter EX's lighting model is very flexible and allows developers to easily add new lights. Existing lights include:

Point Light

The point light is a single point that emits light equally in all directions.

Spotlight

Similar to a flashlight, the spotlight projects light within a specified field of view. The spotlight can also use a texture to tint the color of the lighting on a per pixel basis.

Cube Projector

Similar to the point light, the cube projector uses a cubic texture to tint each lit pixel.

Directional Light

This lighting is emitted from a rectangular plane and is used to simulate directional lights like sunlight.

Point Fill

Although similar to the point light, the point fill is an efficient option because it does not utilize specular lighting or cast shadows.

VISUAL EFFECTS

Jupiter EX features a new optimized, data driven effects system that allows for the creation of key-framed effects that can be comprised of dynamic lights, particle systems, models, and sounds. Examples of the effects that can be created using this system include weapon muzzle flashes, explosions, footsteps, fire, snow, steam, smoke, dust, and debris.



HAVOK PHYSICS

EX includes Havok Physics for models, characters, world geometry, and player movement.



CHARACTER DYNAMICS

With Havok Physics, game characters react naturally in physical environments. In addition to giving Jupiter EX-powered games a more realistic appearance, Havok frees the development team from animating a large number of responses for each collision event.

Jupiter EX includes the following systems from the Havok Character Dynamics Kit:

Physics Shapes

A model node can be encapsulated by a sphere, box, or capsule for use in collision detection.

Constraints

Model nodes can use any of the following constraints: ball and socket, stiff spring, hinge, limited hinge, rag doll, wheel, or prismatic.

Model Animation Blending

Physical effects can be applied to models allowing for dynamic reactions to effects such as getting shot or bumping into a wall.

Rag Doll

Classic rag doll effects blended from key-framed animation sequences for more realistic secondary motions. Plug-ins for rag doll set-up within 3D Studio Max and Maya. This enables developers to easily create, tweak and test character joints within a familiar environment.

Multiple Character Types

Effects for various model types including characters with more than four limbs.

Collision Response Tool

An integrated tool that gives the developer the ability to tune physical responses.

VEHICLE DYNAMICS

In addition to Havok's character dynamics, Jupiter EX includes the Havok Vehicle Kit, which has been designed to allow all types of vehicle behaviors.

The Vehicle Kit includes:

Core Vehicle Physics

Includes a tuned wheel friction model as well as game specific modules for transmission, engine, suspension, aerodynamics and steering.

Debris

Breakable constraints for post-crash debris.

Various Car Types

Support for realistic racing cars, motorcycles cartoon cars, tanks, monster trucks and articulated trucks and trailers.



ENVIRONMENTAL DYNAMICS

Havok's rigid body dynamics allow developers to include environmental physics in their Jupiter EX-powered games.

VISUALIZING PHYSICS DATA

Engineers can use the Havok Visual Debugger to view physics data as well as analyze and profile the game for optimal performance.



MODELING AND ANIMATION

Jupiter EX contains a flexible animation system designed to create highly realistic characters and animated objects. This module includes support for skinning, animation blending, level of detail, lip-syncing, and Havok physics.

Jupiter EX's facial animation system features phoneme-based lip-syncing that support a wide range of expressions to bring game characters to life.

To utilize the Havok character dynamics, models need to be setup with physical shapes and constraints to enable rag doll effects. As explained in the tools section, ModelEdit provides a robust visualization and editing environment for easy adjustments of model data.



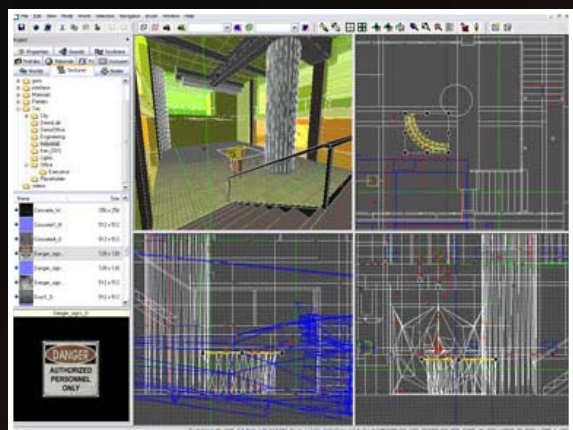
NETWORKING

Jupiter EX's networking module uses a client/server architecture and supports a wide range of multiplayer applications with up to 16 players on each server. Jupiter EX also includes Linux server support.

DATABASE EDITOR

The Game Database Editor (GDE) is a new tool that replaces the Attribute system used in Jupiter and allows easy editing of game attributes and game resources. The GDE also features an optional interface with AlienBrain.

CONTENT TOOLS



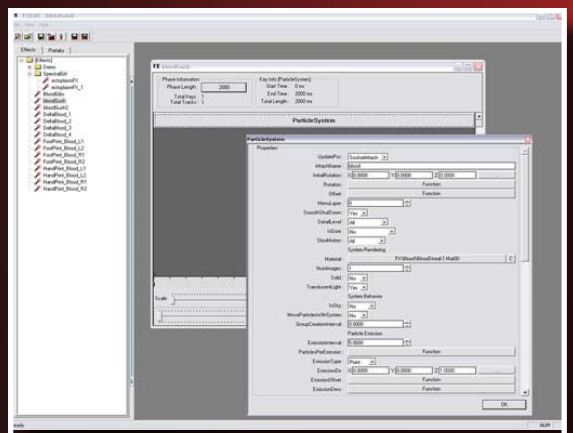
WORLD EDIT

World Edit is similar to the DEdit environment creator in Jupiter with enhancements to support Jupiter EX's new rendering and visibility systems as well as Havok Physics.

World Edit allows level designers to create dynamic 3D environments, worlds, levels, structures and lighting with one easy-to-use tool. The level designer also uses World Edit to populate the world with objects and interactive items.

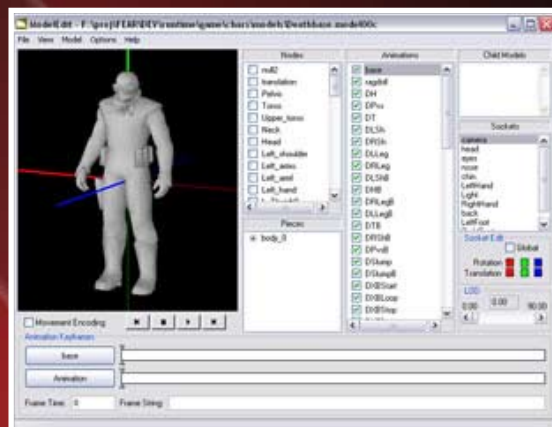
MODEL EDIT

Jupiter EX features the latest version of ModelEdit, which has been extended to include improved UI and changes to support Havok Physics. ModelEdit ensures models are optimized to interact correctly with environments and other objects. Artists can also use ModelEdit to define specific properties, assign texture IDs, attach objects to models and manipulate animation data.



FX EDIT

FxEdit is used to create and modify visual effects. The tool enables the creation of in-game visual effects without requiring a recompile, thereby increasing the efficiency of the design and implementation of special effects. The latest version of FxEdit has been updated with an improved user interface as well as more robust and efficient effects.



LICENSING

Each Jupiter EX license includes complete source code to Jupiter EX's technologies, content creation tools as well as F.E.A.R. game code and content. Licensees also receive:

- Technical support
- Software upgrades
- Content and programming documentation
- Training

Contact sales@touchdownentertainment.com for pricing and additional licensing details.



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